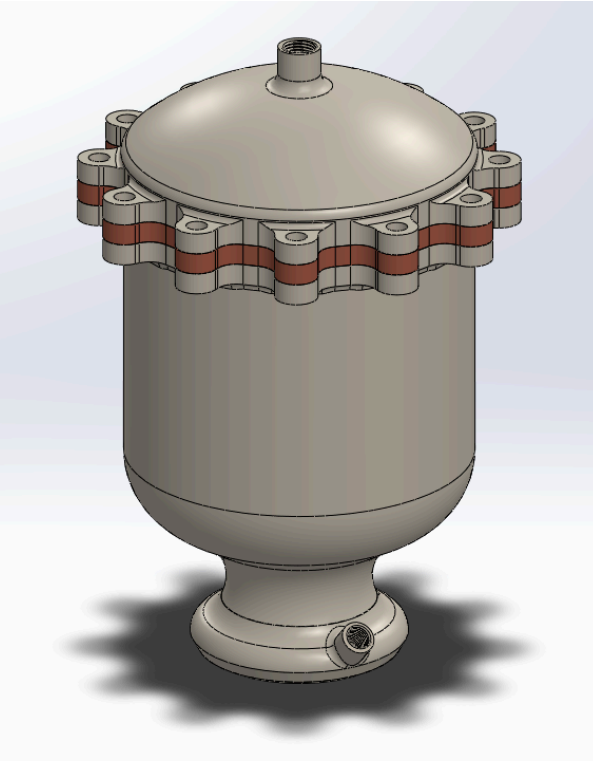
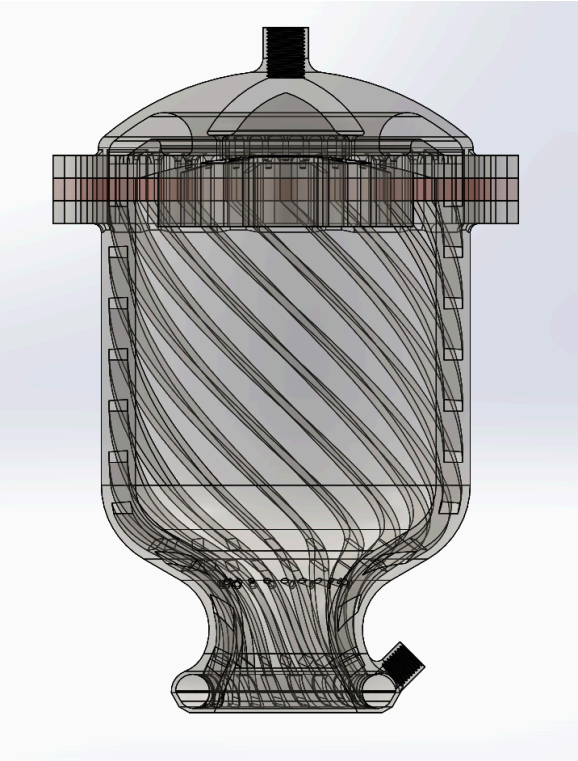
Ram Rocketry - Colorado State University
Liquid Rocket Engine Initial Prototypes & Simulations

Ian Leary & Jay Suthar





Figures 1 & 2: Iteration 1, experimenting with helical internal cooling channel modelling

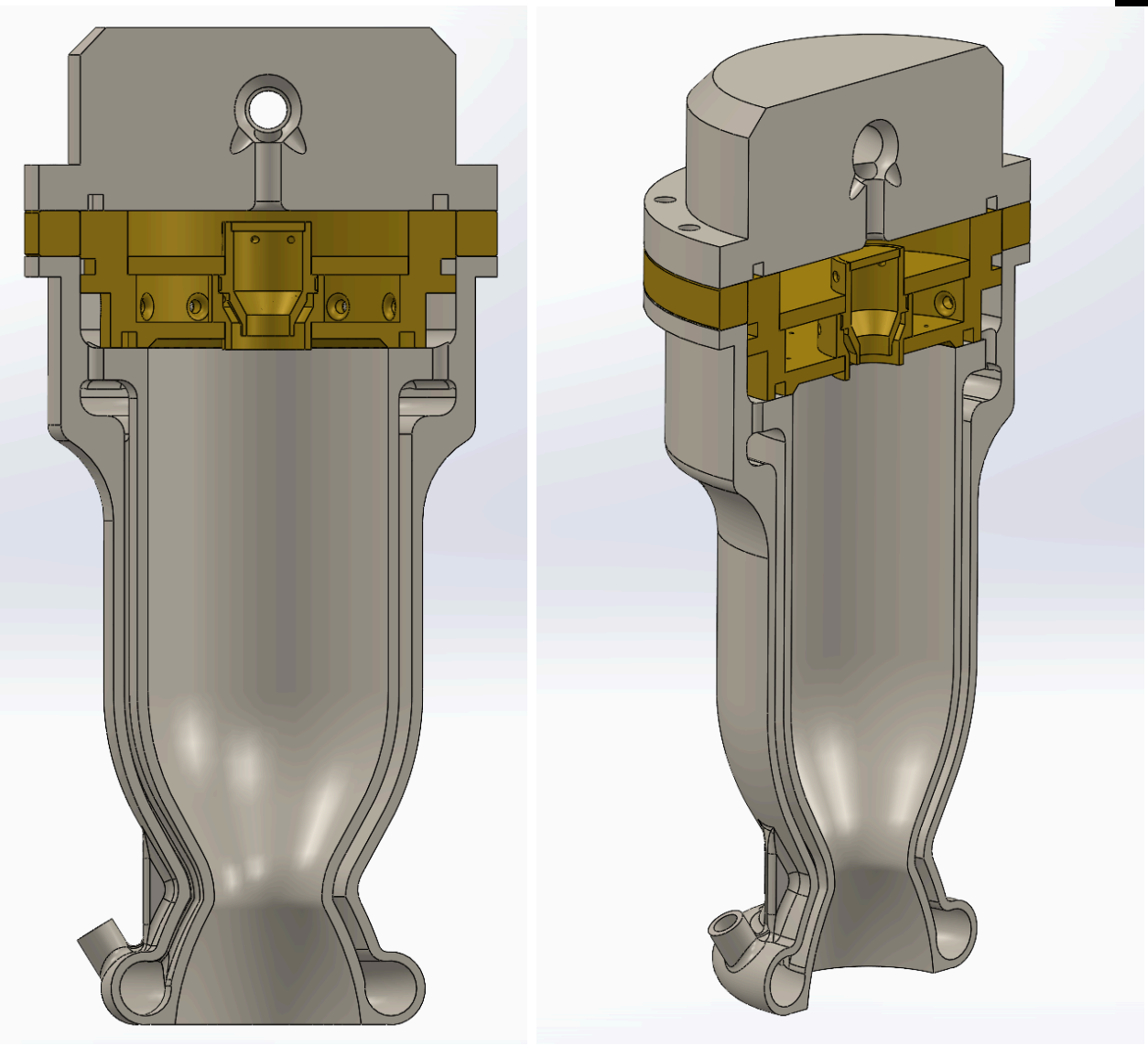


Figure 3 & 4: Iteration 2, Regenerative Cooling Channels, Coaxial Swirl Injector

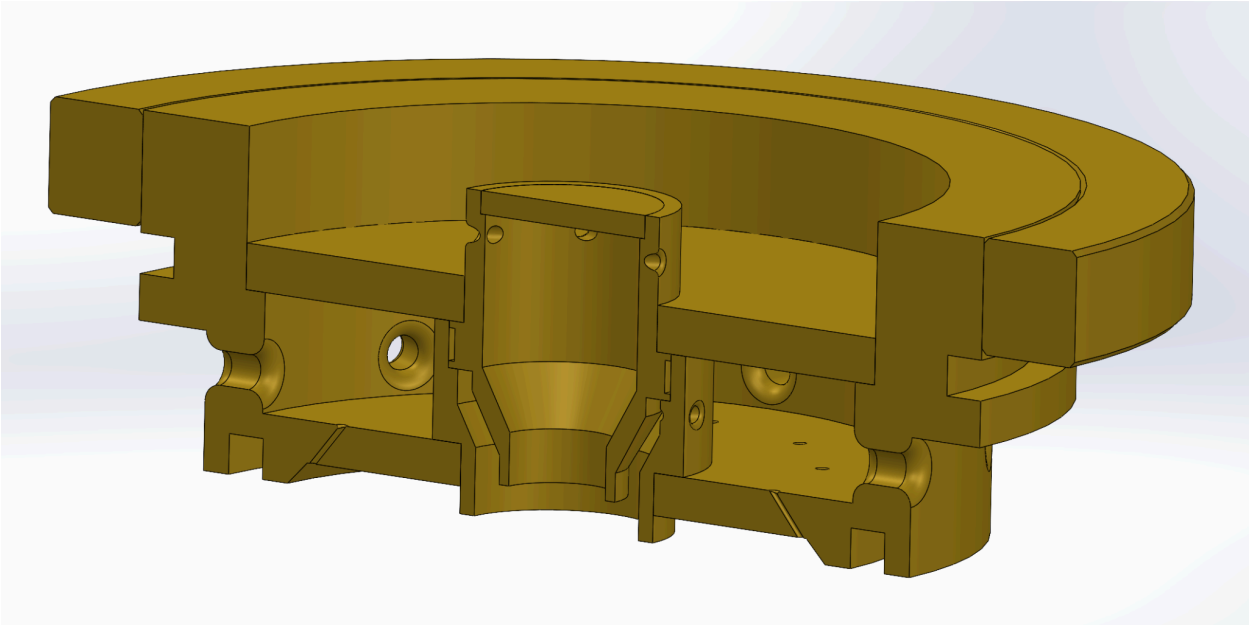


Figure 5: Injector Sub-Assembly, Incorporating Coaxial Swirl Injector and Film Cooling

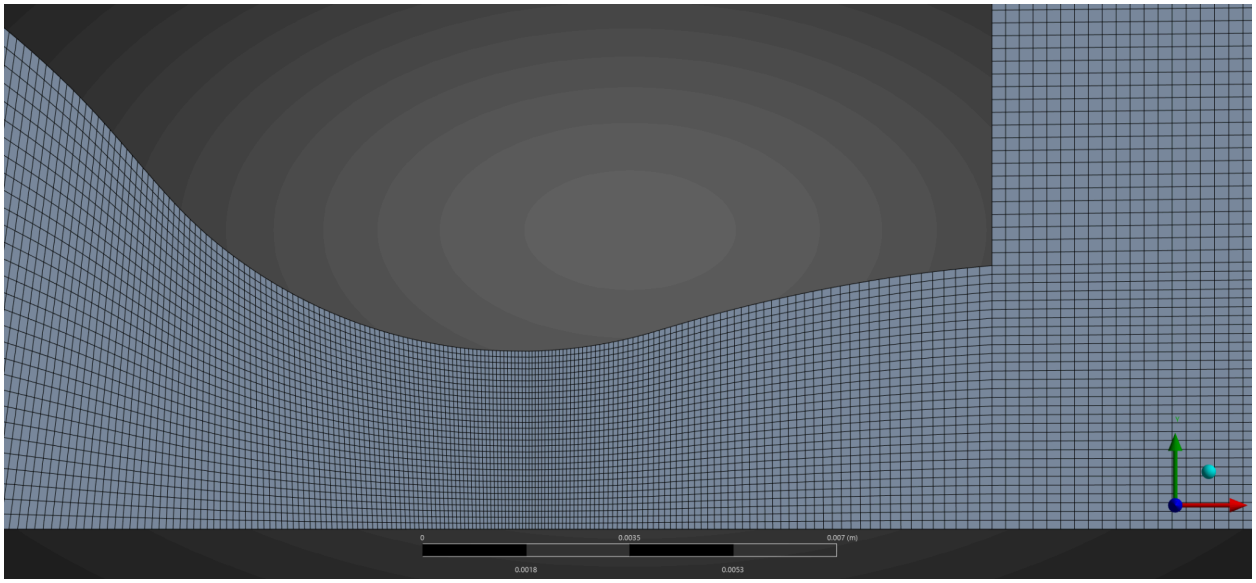


Figure 7: Ansys Mesh View. Highlights nozzle's convergent, divergent, and domain sections

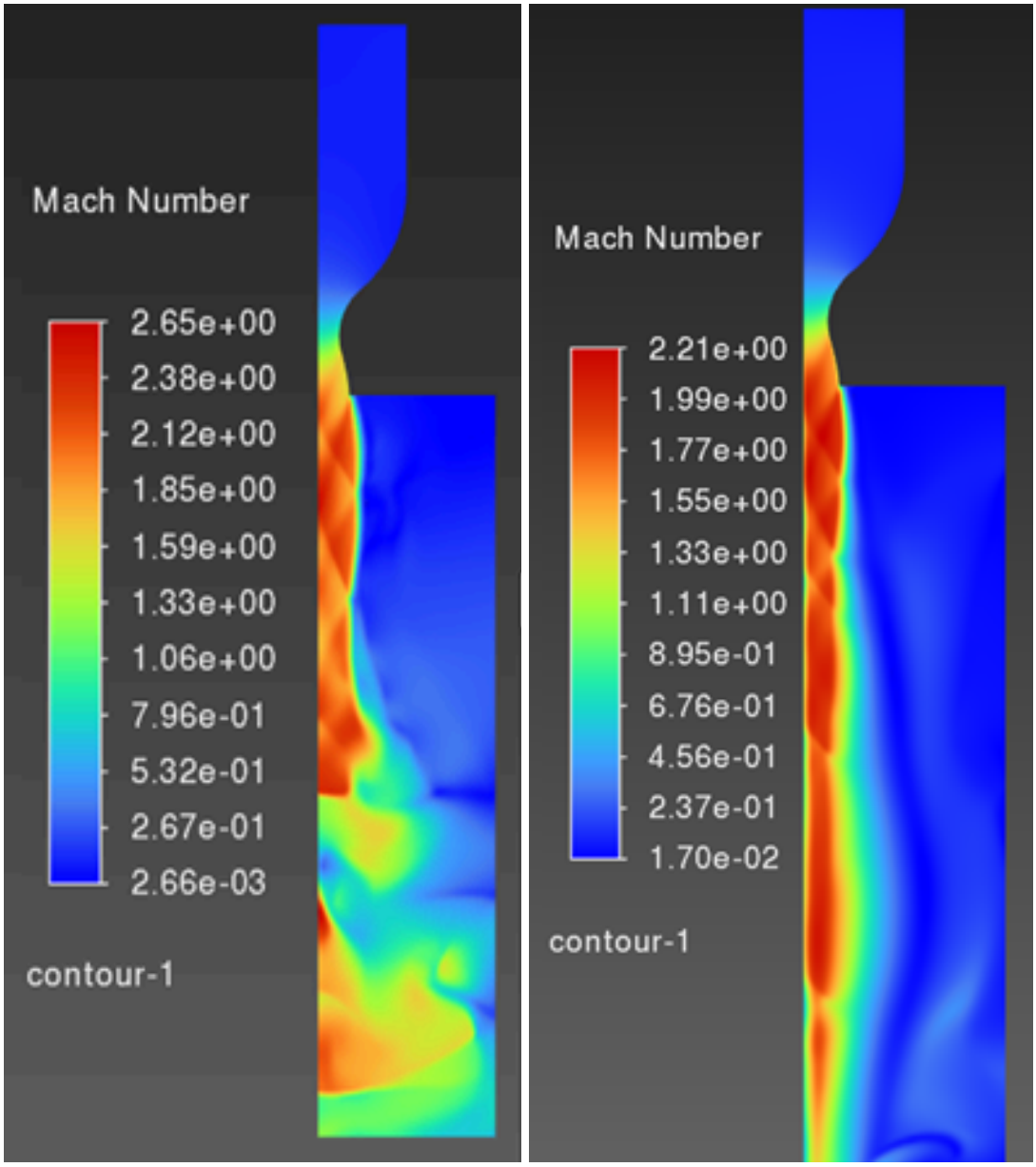


Figure 8 & 9: Nozzle Mach Contour V1 & V2

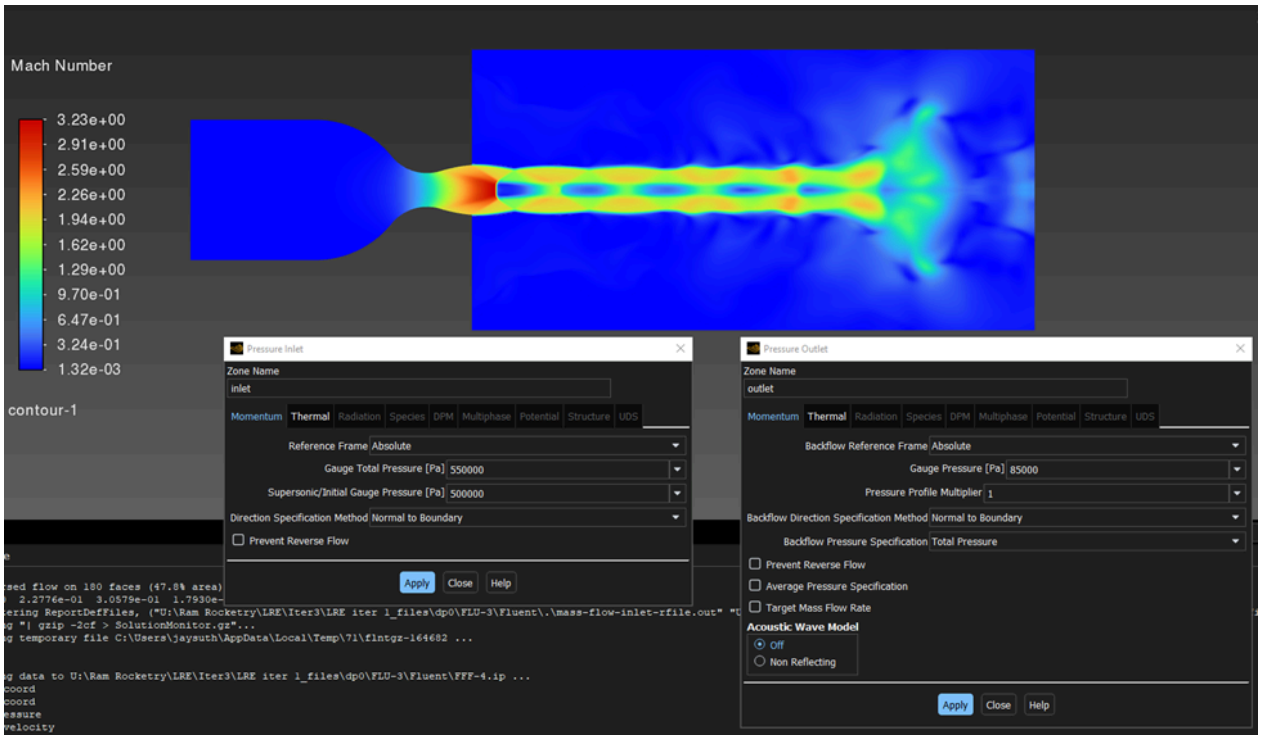


Figure 9: Final mach contour plot of our nozzle geometry with dummy inlet/outlet pressure inputs

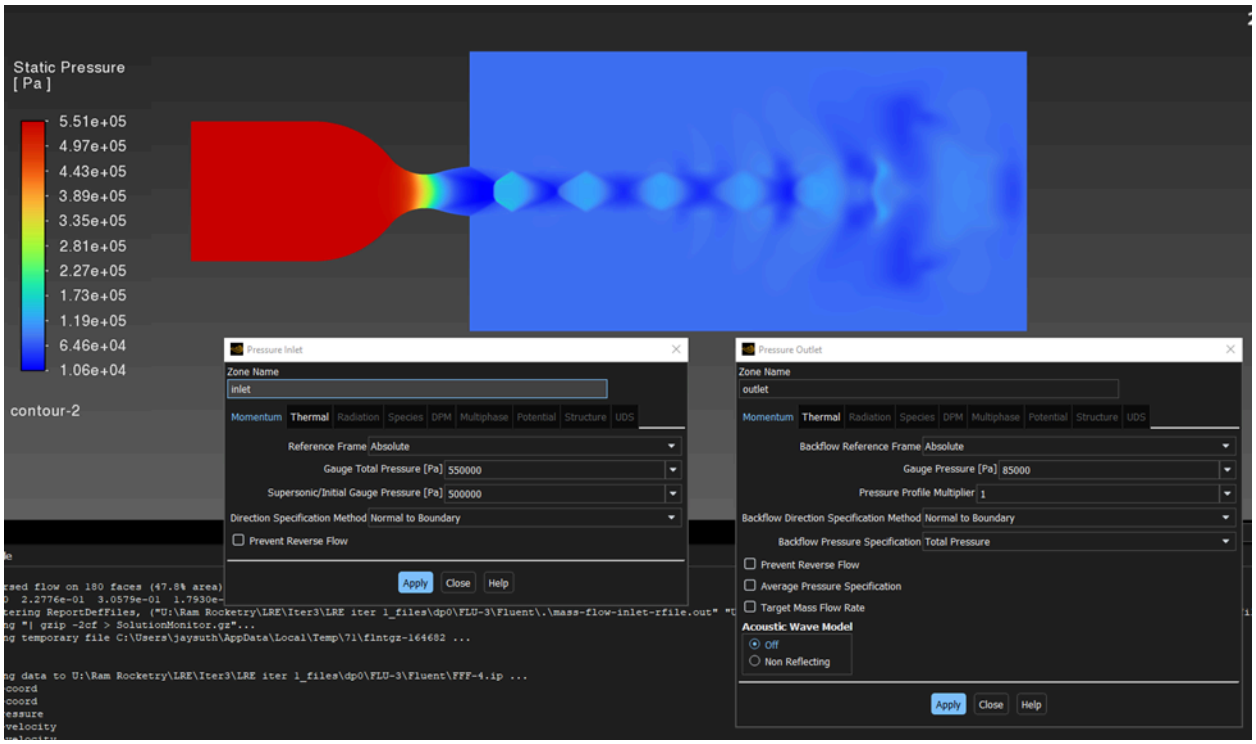


Figure 10: Static Pressure contour plot